WORKED SOLUTION

STEP I 1997, Question 1

Pure • Counting Coin Combinations

STEP I 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q1

This question has a full guided treatment in *Beyond STEP*, Section 9.5, *Making Change*: https://beyondstep.keithmcnulty.org/ch09/making-change.html

The key idea

Counting "ways" only works if we have a system that lists every combination exactly once. The natural system is to fix the number of the largest coin first, then the next largest, and so on. Once only 2p and 1p coins are left, the count is immediate: the number of 2p coins decides everything. Doing it this way also means the 10p answer feeds straight into the 20p answer.

Solution

Order doesn't matter: a "way" is a choice of how many of each coin to use. So we count solutions in non-negative integers.

The basic building block: 2p and 1p coins

To make r pence from 2p and 1p coins, we choose the number b of 2p coins, which can be anything from 0 to $\lfloor r/2 \rfloor$. The 1p coins then make up the remaining $r - 2b \geq 0$ pence, in exactly one way. So the number of ways is

$$\left\lfloor \frac{r}{2} \right\rfloor + 1.$$

Making 10 pence

- *Using a 10p coin.* It makes the whole amount alone: 1 way.
- *No 10p coin.* We split by the number of 5p coins, which is 0, 1 or 2:

5p coins	left for 2p and 1p	ways $\lfloor r/2 \rfloor + 1$
2	0	1
1	5	3
0	10	6

That gives $1 + 3 + 6 = 10$ ways.

Every combination has a definite number of 10p coins and of 5p coins, so it is counted once and only once. The total is $1 + 10 = 11$ ways, as required.

Making 20 pence

Write $F(n)$ for the number of ways to make n pence from 5p, 2p and 1p coins. We have just seen that $F(10) = 10$, and clearly $F(0) = 1$ (use no coins). For $F(20)$ we split by the number of 5p coins, from 0 to 4:

5p coins	left for 2p and 1p	ways
4	0	1
3	5	3
2	10	6
1	15	8
0	20	11

so $F(20) = 1 + 3 + 6 + 8 + 11 = 29$.

Ask yourself How do the 20p and 10p coins fit in without double counting?

Split again by the large coins. Either we use the 20p coin, which must then be the only coin: 1 way. Or we don't, and we split by the number of 10p coins:

- two 10p coins: nothing left, $F(0) = 1$ way;
- one 10p coin: 10p left, $F(10) = 10$ ways;
- no 10p coin: 20p left, $F(20) = 29$ ways.

These cases are disjoint and cover everything, so the number of ways to make 20 pence is

$$1 + 1 + 10 + 29 = 41.$$

Checks

- List the 10p case by hand: 10; 5+5; 5+2+2+1, 5+2+1+1+1, 5+1 × 5; then 2 × 5, 2 × 4+1 × 2, ..., 1 × 10 (six of these). That's 11.
- A short computer count (looping over the number of each coin) gives 11 and 41. It also confirms the intermediate values $F(10) = 10$ and $F(20) = 29$, and that without the 20p coin there are 40 ways.

- Small cases are easy to check by hand: there are $F(5) = 4$ ways to make 5p from 5p, 2p and 1p coins, matching $1 + 3$.

Common slips

- Counting orders, so that 5+2+2+1 and 2+5+1+2 become different "ways". They are the same collection of coins.
- Forgetting the single-coin ways: the lone 10p coin, and the lone 20p coin.
- Giving a bare total with no visible system. The question warns that unexplained answers get no credit, so the case split itself is the point.

Beyond STEP

These are *partitions* of n into parts from a fixed set. The neat way to handle them is a generating function: the number of ways is the coefficient of x^n in

$$\frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})(1-x^{20})},$$

and our case split is just multiplying out these factors one at a time. Generating functions turn counting problems into algebra, and they lead on to Euler's and Ramanujan's work on partitions. A good place to explore this is Siklos, *Advanced Problems in Mathematics*; for the number-theory side, Davenport's *The Higher Arithmetic*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.